Exercise C - II

SUBJECTIVE / INTEGER TYPE QUESTIONS

1. If the equations $a(y + z) = x$, $b(z + x) = y$, $c(x + y) = z$ have nontrivial solutions, then find the value of $\frac{1}{1+a}+\frac{1}{1+b}+\frac{1}{1+c}$.

यदि समीकरणों $a(y + z) = x$, $b(z + x) = y$, $c(x + y) = z$ के अनिरर्थक हल हैं, तो $\frac{1}{1+a}+\frac{1}{1+b}+\frac{1}{1+c}$ का मान ज्ञात कीजिये।

Ans. 2

Sol. $x - ay - az = 0$

$bx - y + bz = 0$

$cx + cy - z = 0$

For non-trivial solution, $D = 0$

$$\begin{vmatrix} 1 & -a & -a \\ b & -1 & b \\ c & c & -1 \end{vmatrix} = 0$$

$$C_1 \to C_1 - C_2$$

$$\begin{vmatrix} 1+a & 0 & -a \\ b+1 & -(1+b) & b \\ 0 & c+1 & -1 \end{vmatrix} = 0$$

$(1 + a)((1 + b) - b(c + 1)) - (1 + b)(a(c + 1)) = 0$

Divide by $(1 + a)(1 + b)(1 + c)$

$$\frac{1}{1+c}-\frac{b}{1+b}-\frac{a}{1+a}=0$$

$$\frac{1}{1+c}+\frac{1}{1+b}+\frac{1}{1+a}=2$$

2. Given $a=\frac{x}{y-z}; b=\frac{y}{z-x}; c=\frac{z}{x-y}$ where x, y, z are not all zero, prove that: $1 + ab + bc + ca = 0$.

दिया गया है $a=\frac{x}{y-z}; b=\frac{y}{z-x}; c=\frac{z}{x-y}$ जहाँ x, y, z सभी शून्य नहीं है, सिद्ध कीजिये कि $1 + ab + bc + ca = 0$.

Ans.

Sol. $ay - az = x, bz - bx = y, cx - cy = z$

$$\begin{vmatrix} 1 & -a & a \\ b & 1 & -b \\ c & -c & -1 \end{vmatrix} = 0$$

$\Rightarrow (-1 - bc) + a(-b + bc) + a(-bc - c) = 0$

$-1 - bc - ab + abc - abc - ac = 0$

$\Rightarrow 1 + ab + bc + ca = 0$

3. If $\sin q \neq \cos q$ and x, y, z satisfy the equations

$x \cos p - y \sin p + z = \cos q + 1$

$x \sin p + y \cos p + z = 1 - \sin q$

$x \cos(p + q) - y \sin (p + q) + z = 2$

then find the value of $x^2 + y^2 + z^2$.

यदि $\sin q \neq \cos q$ तथा x, y, z समीकरणों

$x \cos p - y \sin p + z = \cos q + 1$

$x \sin p + y \cos p + z = 1 - \sin q$

$x \cos(p + q) - y \sin (p + q) + z = 2$

को संतुष्ट करते हैं, तो $x^2 + y^2 + z^2$ का मान ज्ञात कीजियें।

Ans. 2

Sol. $x \cos p - y \sin p + z = \cos q + 1$ _____(1)

$x \sin p + y \cos p + z = 1 - \sin q$ _____(2)

Multiply equation (1) by cos q & equation (2) by sin q & subtracting

$x \cos (p + q) - y \sin (p + q) + z (\cos q - \sin q) = 1 + \cos q - \sin q$ _____(3)

$x \cos (p + q) - y \sin (p + q) + z = 2$ _____(4)

Eq. (3) – (4)

$z(\cos q - \sin q - 1) = -1 + \cos q - \sin q$

$\Rightarrow x \cos p + y \sin p = \cos q$

and $x \sin p + y \cos p = -\sin q$

On solving, $x = \cos(p + q)$

$y = -\sin(p + q)$

$z = 1$

$x^2 + y^2 + z^2 = 2$

4. Investigate for what values of λ, μ the simultaneous equations $x + y + z = 6$; $x + 2y + 3z = 10$ & $x + 2y + \lambda z = \mu$ have :
(a) A unique solution
(b) An infinite number of solutions
(c) No solution

जाँच कीजिये कि λ, μ के किन मानों के लिये युगपत समीकरणों $x + y + z = 6$; $x + 2y + 3z = 10$ तथा $x + 2y + \lambda z = \mu$ के निकाय :
(a) के अद्वितीय हल होंगे
(b) के अन्नत हल होंगे
(c) का कोई हल नहीं होगा

Ans. (a) $\lambda \neq 3$ (b) $\lambda = 3, \mu = 10$ (c) $\lambda = 3, \mu \neq 10$

Sol. $D = \begin{vmatrix} k & 2 & -2 \\ 4 & 2k & -1 \\ 6 & 6 & k \end{vmatrix} = k(2k^2 + 6) - 2(4k + 6) - 2(24 - 12k)$

$\Rightarrow 2k^3 + 6k - 8k - 12 - 48 + 24k$

$\Rightarrow 2k^3 + 22k - 60$

$\Rightarrow 2(k^3 + 11k - 30)$

$\Rightarrow 2(k - 2)(k^2 + 2k + 15)$

For unique solution, $k \neq 2$

$D_1 = \begin{vmatrix} 1 & 2 & -2 \\ 2 & 2k & -1 \\ 3 & 6 & k \end{vmatrix} = (2k^2 + 6) - 2(2k + 3) - 2(12 - 6k)$

$\Rightarrow 2k^2 + 6 - 4k - 6 - 24 + 12k$

$\Rightarrow 2k^2 + 8k - 24$

$\Rightarrow 2(k^2 + 4k - 12) \Rightarrow 2(k + 6)(k - 2)$

$$x = \frac{D_1}{D} = \frac{k+6}{k^2+2k+15}$$

$$D_2 = \begin{vmatrix} k & 1 & -2 \\ 4 & 2 & -1 \\ 6 & 3 & k \end{vmatrix} = k(2k + 3) - (4k + 6) - 2(12 - 12)$$

$\Rightarrow 2k^2 - k - 6$

$\Rightarrow (2k + 3)(k - 2)$

$$y = \frac{2k+3}{2\left(k^2+2k+15\right)}$$

$$D_3 = \begin{vmatrix} k & 2 & 1 \\ 4 & 2k & 2 \\ 6 & 6 & 3 \end{vmatrix} = k(6k - 12) - 2(12 - 12) + (24 - 12k)$$

$\Rightarrow 6k^2 - 12k + 24 - 12k$

$\Rightarrow 6(k^2 - 4k + 4) \Rightarrow 6(k - 2)^2$

$$z = \frac{3(k-2)}{k^2+2k+15}$$

If $k = 2$, infinite solution

Put $x = \lambda$,

we get $y = \frac{1-2\lambda}{2}$, $z = 0$

5. Solve the equations : $Kx + 2y - 2z = 1,\ 4x + 2Ky - z = 2,\ 6x + 6y + Kz = 3$

considering specially the case when $K = 2$. If $K = 2$, then $x = \lambda$, $y = \frac{1-2\lambda}{2}$ and $z = 0$, where $\lambda \in R$

समीकरणों : $Kx + 2y - 2z = 1,\ 4x + 2Ky - z = 2,\ 6x + 6y + Kz = 3$ को

$K = 2$ के विशिष्ट मान के लिये हल कीजियें। यदि $K = 2$, तो $x = \lambda$, $y = \frac{1-2\lambda}{2}$ तथा $z = 0$ जहाँ $\lambda \in R$

Ans. If $K \neq 2$, $\frac{x}{2(K+6)} = \frac{y}{2K+3} = \frac{z}{6(K-2)} = \frac{1}{2\left(K^2+2K+15\right)}$, If K= 2, then $x = \lambda$, $y = \frac{1-2\lambda}{2}$ and $z = 0$ where $\lambda \in R$

Sol. $D = \begin{vmatrix} k & 2 & -2 \\ 4 & 2k & -1 \\ 6 & 6 & k \end{vmatrix} = k(2k^2 + 6) - 2(4k + 6) - 2(24 - 12k)$

$\Rightarrow 2k^3 + 6k - 8k - 12 - 48 + 24k$

$\Rightarrow 2k^3 + 22k - 60$

$\Rightarrow 2(k^3 + 11k - 30)$

$\Rightarrow 2(k - 2)(k^2 + 2k + 15)$

For unique solution, $k \neq 2$

$D_1 = \begin{vmatrix} 1 & 2 & -2 \\ 2 & 2k & -1 \\ 3 & 6 & k \end{vmatrix} = (2k^2 + 6) - 2(2k + 3) - 2(12 - 6k)$

$\Rightarrow 2k^2 + 6 - 4k - 6 - 24 + 12k$

$\Rightarrow 2k^2 + 8k - 24$

$\Rightarrow 2(k^2 + 4k - 12) \Rightarrow 2(k + 6)(k - 2)$

$x = \frac{D_1}{D} = \frac{k+6}{k^2+2k+15}$

$D_2 = \begin{vmatrix} k & 1 & -2 \\ 4 & 2 & -1 \\ 6 & 3 & k \end{vmatrix} = k(2k + 3) - (4k + 6) - 2(12 - 12)$

$\Rightarrow 2k^2 - k - 6$

$\Rightarrow (2k + 3)(k - 2)$

$y = \frac{2k+3}{2\left(k^2+2k+15\right)}$

$D_3 = \begin{vmatrix} k & 2 & 1 \\ 4 & 2k & 2 \\ 6 & 6 & 3 \end{vmatrix} = k(6k - 12) - 2(12 - 12) + (24 - 12k)$

$\Rightarrow 6k^2 - 12k + 24 - 12k$

$\Rightarrow 6(k^2 - 4k + 4) \Rightarrow 6(k-2)^2$

$$z = \frac{3(k-2)}{k^2 + 2k + 15}$$

If $k = 2$, infinite solution

Put $x = \lambda$,

we get $y = \frac{1 - 2\lambda}{2}$, $z = 0$

6. Let a, b, c, d are distinct numbers to be chosen from the set {1, 2, 3, 4, 5}. If the least possible positive solution for x to the system of equations $\left.\begin{matrix} ax + by = 1 \\ cx + dy = 2 \end{matrix}\right\}$ can be expressed in the form $\frac{p}{q}$ where p and q are relatively prime, then find the value of (p + q).

समुच्चय {1, 2, 3, 4, 5} में से चार भिन्न संख्याओं a, b, c, d का चुनाव किया जाता है। यदि समीकरणों के निकाय $\left.\begin{matrix} ax + by = 1 \\ cx + dy = 2 \end{matrix}\right\}$

के न्यूनतम संभावित धनात्मक हल x को $\frac{p}{q}$ के रूप में प्रदर्शित किया जा सकता है, जहाँ *p* तथा *q* सापेक्षतः अभाज्य है, तो (p + q) का मान ज्ञात कीजियें।

Ans. 19

Sol. $$x = \frac{D_1}{D} = \frac{\begin{vmatrix} 1 & b \\ 2 & d \end{vmatrix}}{\begin{vmatrix} a & b \\ c & d \end{vmatrix}} = \frac{d - 2b}{ad - cb}$$

To get least value, d – 2b is minimum & ad – cb will be maximum.

d – 2b = 1

Case-I d = 5, b = 2

$$x = \frac{1}{5a - 2c}$$

5a – 2c should be maximum.

a = 4, c = 1

Case-II d = 3, b = 1

$$x = \frac{1}{3a - c}$$

3a – c should be maximum.

a = 5, c = 2

$x = \frac{1}{18}$ $x = \frac{1}{13}$

$\frac{1}{18} < \frac{1}{13}$

$x = \frac{1}{18} = \frac{p}{q}$, $p + q = 19$

7. Show that the system of equations

$3x - y + 4z = 3$, $x + 2y - 3z = -2$ and $6x + 5y + \lambda z = -3$

has atleast one solution for any real number λ. Find the set of solutions of $\lambda = -5$.

सिद्ध कीजिये कि समीकरणों के निकाय $3x - y + 4z = 3$, $x + 2y - 3z = -2$ तथा $6x + 5y + \lambda z = -3$ का λ के किसी भी वास्तविक मान के लिये कम से कम एक हल है तथा $\lambda = -5$ के लिए हलों का समुच्चय भी ज्ञात कीजियें।

Ans. If $\lambda \neq -5$ then $x = \frac{4}{7}$; $y = -\frac{9}{7}$ and $z = 0$; If $\lambda = 5$ then $x = \frac{4 - 5K}{7}$; $y = \frac{13K - 9}{7}$ and $z = K$ where $K \in R$

Sol. $D = \begin{vmatrix} 3 & -1 & 4 \\ 1 & 2 & -3 \\ 6 & 5 & \lambda \end{vmatrix} = 3(2\lambda + 15) + (\lambda + 18) + 4(5 - 12)$

$\Rightarrow 7\lambda + 45 + 18 - 28$

$\Rightarrow 7(\lambda + 5)$

for $\lambda = -5$

Put $z = k$

$3x - y = 3 - 4k$ _____(1)

$x + 2y = -2 + 3k$ _____(2)

Multiply eq. (1) by 2 and subtract eq. (2)

$7x = 6 - 8k - 2 + 3k$

$x = \frac{4 - 5k}{7}$, $y = \frac{13k - 9}{7}$

8. Solve the system of equations; $\left.\begin{aligned} z+ay+a^2x+a^3=0 \\ z+by+b^2x+b^3=0 \\ z+cy+c^2x+c^3=0 \end{aligned}\right]$

समीकरणों के निकाय $\left.\begin{aligned} z+ay+a^2x+a^3=0 \\ z+by+b^2x+b^3=0 \\ z+cy+c^2x+c^3=0 \end{aligned}\right]$ को सरल कीजिये।

Ans. $x=-(a+b+c)$, $y=ab+bc+ca$, $z=-abc$

Sol. $D=\begin{vmatrix} a^2 & a & 1 \\ b^2 & b & 1 \\ c^2 & c & 1 \end{vmatrix}=(a-b)(b-c)(c-a)$

$$D_1=\begin{vmatrix} -a^3 & a & 1 \\ -b^3 & b & 1 \\ -c^3 & c & 1 \end{vmatrix}=-(a-b)(b-c)(c-a)(a+b+c)$$

$$D_2=\begin{vmatrix} a^2 & -a^3 & 1 \\ b^2 & -b^3 & 1 \\ c^2 & -c^3 & 1 \end{vmatrix}=-(a-b)(b-c)(c-a)(ab+bc+ca)$$

$$D_3=\begin{vmatrix} a^2 & a & -a^3 \\ b^2 & b & -b^3 \\ c^2 & c & -c^3 \end{vmatrix}=-abc(a-b)(b-c)(c-a)$$

$$x=\frac{D_1}{D}=-(a+b+c)$$

$$y=\frac{D_2}{D}=-(ab+bc+ca)$$

$$z = \frac{D_3}{D} = -abc$$

9. If $a^2 + b^2 + c^2 = 1$ then show that the value of the determinant

$$\begin{vmatrix} a^2+(b^2+c^2)\cos\theta & ba(1-\cos\theta) & ca(1-\cos\theta) \\ ab(1-\cos\theta) & b^2+(c^2+a^2)\cos\theta & cb(1-\cos\theta) \\ ac(1-\cos\theta) & bc(1-\cos\theta) & c^2+(a^2+b^2)\cos\theta \end{vmatrix}$$ simplifies to $\cos^2\theta$.

यदि $a^2 + b^2 + c^2 = 1$ तो सिद्ध कीजिये सारणिक $\begin{vmatrix} a^2+(b^2+c^2)\cos\theta & ba(1-\cos\theta) & ca(1-\cos\theta) \\ ab(1-\cos\theta) & b^2+(c^2+a^2)\cos\theta & cb(1-\cos\theta) \\ ac(1-\cos\theta) & bc(1-\cos\theta) & c^2+(a^2+b^2)\cos\theta \end{vmatrix}$ का

मान $\cos^2\theta$ में सरल हो जाता है।

Ans.

Sol. $a^2 + b^2 + c^2 = 1$

$$\Delta = \begin{vmatrix} a^2+(b^2+c^2)\cos\theta & ab(1-\cos\theta) & ac(1-\cos\theta) \\ ba(1-\cos\theta) & b^2+(c^2+a^2)\cos\theta & bc(1-\cos\theta) \\ ca(1-\cos\theta) & cb(1-\cos\theta) & c^2+(a^2+b^2)\cos\theta \end{vmatrix}$$

$$\Delta = \frac{1}{abc}\begin{vmatrix} a^3+a(b^2+c^2)\cos\theta & b^2a(1-\cos\theta) & ac^2(1-\cos\theta) \\ ba^2(1-\cos\theta) & b^3+b(c^2+a^2)\cos\theta & bc^2(1-\cos\theta) \\ ca^2(1-\cos\theta) & cb^2(1-\cos\theta) & c^3+c(a^2+b^2)\cos\theta \end{vmatrix}$$

$$\frac{abc}{abc}\begin{vmatrix} a^2+(b^2+c^2)\cos\theta & b^2(1-\cos\theta) & c^2(1-\cos\theta) \\ a^2(1-\cos\theta) & b^2+(c^2+a^2)\cos\theta & c^2(1-\cos\theta) \\ a^2(1-\cos\theta) & b^2(1-\cos\theta) & c^2+(a^2+b^2)\cos\theta \end{vmatrix}$$

$C_1 \to C_1 + C_2 + C_3$

$$= (a^2 + b^2 + c^2)\begin{vmatrix} 1 & b^2(1-\cos\theta) & c^2(1-\cos\theta) \\ 1 & b^2+(c^2+a^2)\cos\theta & c^2(1-\cos\theta) \\ 1 & b^2(1-\cos\theta) & c^2+(a^2+b^2)\cos\theta \end{vmatrix}$$

Applying $R_2 \to R_2 - R_1$ and $R_3 \to R_3 - R_1$

संक्रिया $R_2 \to R_2 - R_1$ एवं $R_3 \to R_3 - R_1$ से

$$(a^2 + b^2 + c^2)\begin{vmatrix} 1 & b^2(1-\cos\theta) & c^2(1-\cos\theta) \\ 0 & (a^2+b^2+c^2)\cos\theta & 0 \\ 0 & 0 & (a^2+b^2+c^2)\cos\theta \end{vmatrix} = (a^2 + b^2 + c^2)^3 \cos^2\theta = \cos^2\theta$$

This is independent from a, b, c. यह a, b, c से स्वतन्त्र है।

10. In a Δ ABC, determine condition under which $\begin{vmatrix} \cot\frac{A}{2} & \cot\frac{B}{2} & \cot\frac{C}{2} \\ \tan\frac{B}{2}+\tan\frac{C}{2} & \tan\frac{C}{2}+\tan\frac{A}{2} & \tan\frac{A}{2}+\tan\frac{B}{2} \\ 1 & 1 & 1 \end{vmatrix} = 0.$

एक Δ ABC में, वह प्रतिबन्ध ज्ञात कीजिये जिसके लिये $\begin{vmatrix} \cot\frac{A}{2} & \cot\frac{B}{2} & \cot\frac{C}{2} \\ \tan\frac{B}{2}+\tan\frac{C}{2} & \tan\frac{C}{2}+\tan\frac{A}{2} & \tan\frac{A}{2}+\tan\frac{B}{2} \\ 1 & 1 & 1 \end{vmatrix} = 0$

Ans. Triangle ABC is isosceles

Sol. $C_1 \to C_1 - C_2$, $C_2 \to C_2 - C_3$

$$\begin{vmatrix} \cot\frac{A}{2}-\cot\frac{B}{2} & \cot\frac{B}{2}-\cot\frac{C}{2} & \cot\frac{C}{2} \\ \tan\frac{B}{2}-\tan\frac{A}{2} & \tan\frac{C}{2}-\tan\frac{B}{2} & \tan\frac{A}{2}+\tan\frac{B}{2} \\ 0 & 0 & 1 \end{vmatrix} = 0$$

$$\left(\cot\frac{A}{2}-\cot\frac{B}{2}\right)\left(\cot\frac{B}{2}-\cot\frac{C}{2}\right)\begin{vmatrix} 1 & 1 & \cot\frac{C}{2} \\ \frac{1}{\cot\frac{A}{2}\cot\frac{B}{2}} & \frac{1}{\cot\frac{B}{2}\cot\frac{C}{2}} & \tan\frac{A}{2}+\tan\frac{B}{2} \\ 0 & 0 & 1 \end{vmatrix}=0$$

$A = B$ or $B = C$ or $C = A$

ΔABC is an isosceles triangle.

11. If $ax_1^2 + by_1^2 + cz_1^2 = ax_2^2 + by_2^2 + cz_2^2 = ax_3^2 + by_3^2 + cz_3^2 = d$
and $ax_2x_3 + by_2y_3 + cz_2z_3 = ax_3x_1 + by_3y_1 + cz_3z_1 = ax_1x_2 + by_1y_2 + cz_1z_2 = f$,

then prove that $\begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix} = (d-f)\left[\frac{d+2f}{abc}\right]^{\frac{1}{2}}$ $(a, b, c \neq 0)$

यदि $ax_1^2 + by_1^2 + cz_1^2 = ax_2^2 + by_2^2 + cz_2^2 = ax_3^2 + by_3^2 + cz_3^2 = d$
तथा $ax_2x_3 + by_2y_3 + cz_2z_3 = ax_3x_1 + by_3y_1 + cz_3z_1 = ax_1x_2 + by_1y_2 + cz_1z_2 = f$,

तो सिद्ध कीजिए $\begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix} = (d-f)\left[\frac{d+2f}{abc}\right]^{\frac{1}{2}}$ $(a, b, c \neq 0)$

Ans.

Sol. $\Delta = \begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix}$

$$\Delta^2 = \frac{1}{abc}\begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix} \times \begin{vmatrix} ax_1 & by_1 & cz_1 \\ ax_2 & by_2 & cz_3 \\ ax_3 & by_3 & cz_3 \end{vmatrix}$$

$$= \frac{1}{abc}\begin{vmatrix} d & f & f \\ f & d & f \\ f & f & d \end{vmatrix}$$

$C_1 \to C_1 + C_2 + C_3$

$$= \frac{1}{abc}\begin{vmatrix} d+2f & f & f \\ d+2f & d & f \\ d+2f & f & d \end{vmatrix} = \frac{(d+2f)}{abc}\begin{vmatrix} 1 & f & f \\ 1 & d & f \\ 1 & f & d \end{vmatrix}$$

on solving it we get

$$\Delta^2 = \frac{(d+2f)}{abc}(d-f)^2$$

$$\Delta = (d-f)\left[\frac{d+2f}{abc}\right]^{\frac{1}{2}}$$

12. If $u = ax^2 + 2bxy + cy^2$, $u' = a'x^2 + 2b'xy + c'y^2$. Prove that

यदि $u = ax^2 + 2bxy + cy^2$, $u' = a'x^2 + 2b'xy + c'y^2$ हो, तो सिद्ध कीजिये कि

$$\begin{vmatrix} y^2 & -xy & x^2 \\ a & b & c \\ a' & b' & c' \end{vmatrix} = \begin{vmatrix} ax+by & bx+cy \\ a'x+b'y & b'x+c'y \end{vmatrix} = -\frac{1}{y}\begin{vmatrix} u & u' \\ ax+by & a'x+b'y \end{vmatrix}.$$

Ans.

Sol. $\begin{vmatrix} y^2 & -xy & x^2 \\ a & b & c \\ a' & b' & c' \end{vmatrix}$

$$\Rightarrow y^2(bc'-b'c) + xy(ac'-a'c) + x^2(ab'-a'b)$$

$$\begin{vmatrix} ax+by & bx+cy \\ a'x+b'y & b'x+c'y \end{vmatrix} = (ax+by)(b'x+c'y) - (bx+cy)(a'x+b'y)$$

$$\Rightarrow ab'x^2 + ac'xy + bb'xy + bc'y^2 - a'bx^2 - bb'xy - a'cxy - b'cy^2$$

$$\Rightarrow y^2(bc'-b'c) + xy(ac'-a'c) + x^2(ab'-a'b)$$

$$-\frac{1}{y}\begin{vmatrix} ax^2+bxy+cy^2 & a'x^2+2b'xy+c'y^2 \\ ax+by & a'x+b'y \end{vmatrix}$$

$$\Rightarrow -\frac{1}{y}\,(aa'x^3 + ab'x^2y + 2a'bx^2y + 2bb'xy^2 + a'cxy^2 + b'cy^3 - aa'x^3 - 2ab'x^2y$$

$-ac'xy^2 - a'bx^2y - 2b'bxy^2 - bc'y^3)$

$\Rightarrow -ab'x^2 - 2a'bx^2 - a'cxy - b'cy^2 + 2ab'x^2 + ac'xy + a'bx^2 + bc'y^2$

$\Rightarrow x^2(ab' - a'b) + xy(ac' - a'c) + y^2(bc' - b'c)$

13. Test the consistency and solve them when consistent, the following system of equations for all values of λ

λ के मानों के लिये दिये गये निकाय के संगत तथा असंगत होने की जाँच कीजिए तथा प्रत्येक स्थिति में हल भी ज्ञात किजिए।

$x + y + z = 1$

$x + 3y - 2z = \lambda$

$3x + (\lambda + 2)y - 3z = 2\lambda + 1$

Ans. If $\lambda = 5$, system is consistent with infinite solution given by $z = K$, $y = \frac{1}{2}(3K + 4)$ and $x = -\frac{1}{2}(5K + 2)$ where K $\in$ R. If $\lambda \neq 5$, system is consistent with unique solution given by $z = \frac{1}{3}(1 - \lambda)$; $x = \frac{1}{3}(\lambda + 2)$ and $y = 0$.

Sol. $$D = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 3 & -2 \\ 3 & \lambda + 2 & -3 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 0 \\ 1 & 2 & -3 \\ 3 & \lambda - 1 & -6 \end{vmatrix}$$

$\Rightarrow -12 + 3\lambda - 3 \Rightarrow 3\lambda - 15 = 3(\lambda - 5)$

If $\lambda \neq 5$, unique solution

$$D_1 = \begin{vmatrix} 1 & 1 & 1 \\ \lambda & 3 & -2 \\ 2\lambda + 1 & \lambda + 2 & -3 \end{vmatrix} = -9 + 2\lambda + 4 + 3\lambda - 4\lambda - 2 + \lambda^2 + 2\lambda - 6\lambda - 3$$

$\Rightarrow \lambda^2 - 3\lambda - 10 \Rightarrow (\lambda - 5)(\lambda + 2)$

$$D_2 = \begin{vmatrix} 1 & 1 & 1 \\ 1 & \lambda & -2 \\ 3 & 2\lambda + 1 & -3 \end{vmatrix} = -3\lambda + 4\lambda + 2 + 3 - 6 + 2\lambda + 1 - 3\lambda$$

$\Rightarrow 0$

$$D_3 = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 3 & \lambda \\ 3 & \lambda + 2 & 2\lambda + 1 \end{vmatrix} \Rightarrow 6\lambda + 3 - \lambda^2 - 2\lambda - 2\lambda - 1 + 3\lambda + \lambda + 2 - 9$$

$\Rightarrow \lambda^2 + 6\lambda - 5$

$\Rightarrow -(\lambda - 1)(\lambda - 5)$

$$x = \frac{1}{3}(\lambda + 2),\ y = 0,\ z = \frac{-1}{3}(\lambda - 1)$$

If $\lambda = 5$, put $z = k$

$x + y = 1 - k$

$x + 3y = 5 + 2k$

$$y = \frac{1}{2}(3k + 4), x = \frac{-1}{2}(5k + 2)$$

14. Prove that the value of the determinant $\begin{vmatrix} -7 & 5+3i & \frac{2}{3}-4i \\ 5-3i & 8 & 4+5i \\ \frac{2}{3}+4i & 4-5i & 9 \end{vmatrix}$ is real.

सिद्ध कीजिये कि सारणिक $\begin{vmatrix} -7 & 5+3i & \frac{2}{3}-4i \\ 5-3i & 8 & 4+5i \\ \frac{2}{3}+4i & 4-5i & 9 \end{vmatrix}$ का मान वास्तविक है।

Ans.

Sol. Let $A = \begin{vmatrix} -7 & 5+3i & \frac{2}{3}-4i \\ 5-3i & 8 & 4+5i \\ \frac{2}{3}+4i & 4-5i & 9 \end{vmatrix}$

$$\overline{A} = \begin{vmatrix} -7 & 5-3i & \frac{2}{3}+4i \\ 5+3i & 8 & 4-5i \\ \frac{2}{3}-4i & 4+5i & 9 \end{vmatrix} = A^T$$

$A = |\overline{A}| = |A^T|$

|A| is real.

15. Prove that $\begin{vmatrix} 1+a^2-b^2 & 2ab & -2b \\ 2ab & 1-a^2+b^2 & 2a \\ 2b & -2a & 1-a^2-b^2 \end{vmatrix} = (1 + a^2 + b^2)^3$.

सिद्ध कीजिये $\begin{vmatrix} 1+a^2-b^2 & 2ab & -2b \\ 2ab & 1-a^2+b^2 & 2a \\ 2b & -2a & 1-a^2-b^2 \end{vmatrix} = (1 + a^2 + b^2)^3$.

Ans.

Sol. $\begin{vmatrix} 1+a^2-b^2 & 2ab & -2b \\ 2ab & 1-a^2+b^2 & 2a \\ 2b & -2a & 1-a^2-b^2 \end{vmatrix}$

$C_1 \to C_1 - bC_3$, $C_2 \to C_2 + aC_3$

$$= \begin{vmatrix} 1+a^2+b^2 & 0 & -2b \\ 0 & 1+a^2+b^2 & 2a \\ b(1+a^2+b^2) & -a(1+b^2+a^2) & 1-a^2-b^2 \end{vmatrix}$$

$$= (1 + a^2 + b^2)^2 \begin{vmatrix} 1 & 0 & -2b \\ 0 & 1 & 2a \\ b & -a & 1-a^2-b^2 \end{vmatrix}$$

$\equiv (1 + a^2 + b^2)^3$

16. Show that the value of the determinant $\begin{vmatrix} \tan(A+P) & \tan(B+P) & \tan(C+P) \\ \tan(A+Q) & \tan(B+Q) & \tan(C+Q) \\ \tan(A+R) & \tan(B+R) & \tan(C+R) \end{vmatrix}$ vanishes for all values of

A, B, C, P, Q & R where A + B + C + P + Q + R = 0

सिद्ध कीजिये कि सारणिक $\begin{vmatrix} \tan(A+P) & \tan(B+P) & \tan(C+P) \\ \tan(A+Q) & \tan(B+Q) & \tan(C+Q) \\ \tan(A+R) & \tan(B+R) & \tan(C+R) \end{vmatrix}$ का मान A, B, C, P, Q तथा R जहाँ

`A + B + C + P + Q + R = 0 के प्रत्येक सम्भव संचय के लिए शून्य होगा।

Ans.

Sol. If $X + Y + Z = 0$ then $\tan X + \tan Y + \tan Z = \tan X \tan Y \tan Z$

Expanding by R_1 :

$\tan (A + P) (\tan (B + Q) \tan (C + R) - \tan (B + R) \tan (C + Q))$
$- \tan (B + P (\tan (A + Q) \tan (C + R) - \tan (A + R) \tan (C + Q))$
$+ \tan (C + P) (\tan (A + Q) \tan (B + R) - \tan (A + R) \tan (B + Q)) = 0$

COMPREHENSION TYPE QUESTIONS

Paragraph for Question nos. 17 to 19

Let $x, y, z \in R^+$ & $\Delta = \begin{vmatrix} x & x^3 & x^4 - 1 \\ y & y^3 & y^4 - 1 \\ z & z^3 & z^4 - 1 \end{vmatrix}$

माना $x, y, z \in R^+$ तथा $\Delta = \begin{vmatrix} x & x^3 & x^4 - 1 \\ y & y^3 & y^4 - 1 \\ z & z^3 & z^4 - 1 \end{vmatrix}$

17. If $x \neq y \neq z$ & x, y, z are in GP and $\Delta = 0$ then y is equal to :
(A) 1 (B) 2 (C) 4 (D) None of these

यदि $x \neq y \neq z$ तथा x, y, z गुणोतर श्रेणी में है तथा $\Delta = 0$ तो y होगा :
(A) 1 (B) 2 (C) 4 (D) इनमें से कोई नहीं

Ans. A

Sol. Put $x = \frac{y}{r}, z = yr$

$\Rightarrow y = 1$

18. If x, y, z are roots of $t^3 - 21t^2 + bt - 343 = 0$, $b \in R$ then $\Delta =$
(A) 1 (B) 0
(C) depends on x, y, z (D) data inadequate

यदि x, y, z समीकरण $t^3 - 21t^2 + bt - 343 = 0$, $b \in R$ के मूल है, तो Δ =

(A) 1 (B) 0

(C) x, y, z पर निर्भर करता है (D) अपर्याप्त जानकारी

Ans. B

Sol. $\left(\frac{y}{r}\right)(y)(yr) = 343 \Rightarrow y = 7$

$\frac{7}{r} + 7 + 7r = 21$

$\Rightarrow r = 1$ $x = y = z$

$\Delta = 0$

19. If x, y, z are in A.P. and $\Delta = 0$ then $2xy^2z + x^2z^2$ =

(A) 1 (B) 2 (C) 3 (D) None of these

यदि x, y, z A.P. में हो तथा $\Delta = 0$ तो $2xy^2z + x^2z^2$ =

(A) 1 (B) 2 (C) 3 (D) इनमें से कोई नहीं

Ans. C

Sol. $x = y = z = 1 \Rightarrow d = 0$

$2xy^2z + x^2z^2 = 3$

Paragraph for Question nos. 20 to 22

Consider the system of equations $2x + ay + 6z = 8$, $x + 2y + bz = 5$ and $x + y + 3z = 4$ then

माना समीकरण निकाय $2x + ay + 6z = 8$, $x + 2y + bz = 5$ तथा $x + y + 3z = 4$ है, तो

20. The system of equations will have infinite solutions if :

समीकरण निकाय के अनन्त हल होगें, यदि :

(A) $b = 3$ (B) $b = 2$ (C) $a = 3$ (D) $a = 2$

Ans. D

21. The given system of equations will have no solution if :

समीकरण निकाय का कोई हल नहीं होगा, यदि :

(A) $a = 2, b \neq 3$ (B) $a \neq 2, b \neq 3$ (C) $a \neq 2, b = 3$ (D) $a = 2, b = 3$

Ans. C

22. The given system of equations will have a unique solution if :

समीकरण निकाय का अद्वितीय हल होगा, यदि :

(A) $a \neq 2, b \neq 3$ (B) $a \neq 2$ (C) $a = 2, b = 3$ (D) $a = 2, b \neq 3$

Ans. A

Sol. $$D = \begin{vmatrix} 2 & a & 6 \\ 1 & 2 & b \\ 1 & 1 & 3 \end{vmatrix} = 2(6-b) - a(3-b) + 6(1-2)$$

$\Rightarrow 12 - 2b - 3a + ab - 6$

$\Rightarrow a(b-3) - 2(b-3) \Rightarrow (a-2)(b-3)$

$$D_1 = \begin{vmatrix} 8 & a & 6 \\ 5 & 2 & b \\ 4 & 1 & 3 \end{vmatrix} = 8(6-b) - a(15-4b) + 6(5-8)$$

$\Rightarrow 48 - 8b - 15a + 4ab - 18 \Rightarrow 4b(a-2) - 15(a-2)$

$\Rightarrow (4b-15)(a-2)$

$$D_2 = \begin{vmatrix} 2 & 8 & 6 \\ 1 & 5 & b \\ 1 & 4 & 3 \end{vmatrix} = 2(15-4b) - (24-24) + (8b-30)$$

$\Rightarrow 0$

$$D_3 = \begin{vmatrix} 2 & a & 8 \\ 1 & 2 & 5 \\ 1 & 1 & 4 \end{vmatrix} = 2(8-5) - a(4-5) + 8(1-2)$$

$\Rightarrow 6 + a - 8$

$\Rightarrow a - 2$

For unique solution; $D \neq 0$; $a \neq 2, b \neq 3$

For infinitely many solutions; $D = D_1 = D_2 = D_3 = 0$

$\Rightarrow a = 2$

For no solution; $D = 0$ & either one of $D_1, D_2, D_3 \neq 0$

$a \neq 2, b = 3$

Paragraph for Question nos. 23 to 24

Given $f(x)=\begin{vmatrix} 5+\sin^2 x & \cos^2 x & 4\sin 2x \\ \sin^2 x & 5+\cos^2 x & 4\sin 2x \\ \sin^2 x & \cos^2 x & 5+4\sin 2x \end{vmatrix}$, then :

दिया गया है कि $f(x)=\begin{vmatrix} 5+\sin^2 x & \cos^2 x & 4\sin 2x \\ \sin^2 x & 5+\cos^2 x & 4\sin 2x \\ \sin^2 x & \cos^2 x & 5+4\sin 2x \end{vmatrix}$, तो

23. The domain of the function f(x) :

फलन f(x) का प्रांत होगा :

(A) $(-\infty, 0)$ (B) $(-\infty,\infty)$ (C) $(0, \infty)$ (D) (0, 1)

Ans. B

24. The range of the function f(x) :

(A) [50, 250] (B) [25, 250] (C) [50, 50] (D) None of these

फलन f(x) का परिसर होगा :

(A) [50, 250] (B) [25, 250] (C) [50, 50] (D) इनमें से कोई नहीं

Ans. A

Sol. $R_2 \to R_2 - R_1, R_3 \to R_3 - R_1$

$$f(x)=\begin{vmatrix} 5+\sin^2 x & \cos^2 x & 4\sin 2x \\ -5 & 5 & 0 \\ -5 & 0 & 5 \end{vmatrix}$$

$$\Rightarrow 5\left(5\cos^2 x\right)+5\left(25+5\sin^2 x+20\sin 2x\right)$$

$$\Rightarrow 25+125+100\sin 2x$$

$$\Rightarrow 150+100\sin 2x$$

Domain : R

Range : [50, 250]